

TECHNICAL DETAILS

SkinSync

Under the hood: vision, color science, data and security

React + Vite front end, in-browser MediaPipe face landmarks, a CIELAB color pipeline, and Supabase for auth, Postgres with Row Level Security, and storage. All on free tiers.



Technology Stack

Every layer runs on a free tier or open source

Frontend

React 18

Vite

Tailwind CSS

React Router v6

lucide-react

jsPDF

Vision and color

MediaPipe Tasks Vision

Canvas API

Custom sRGB to CIELAB

Monk Skin Tone Scale (CC BY 4.0)

Backend

Supabase Auth

Postgres + RLS

Storage

supabase-js

Hosting and DevOps

Vercel

GitHub

GitHub Actions

jsDelivr / GCS CDN

Quality and design

Vitest

Lighthouse

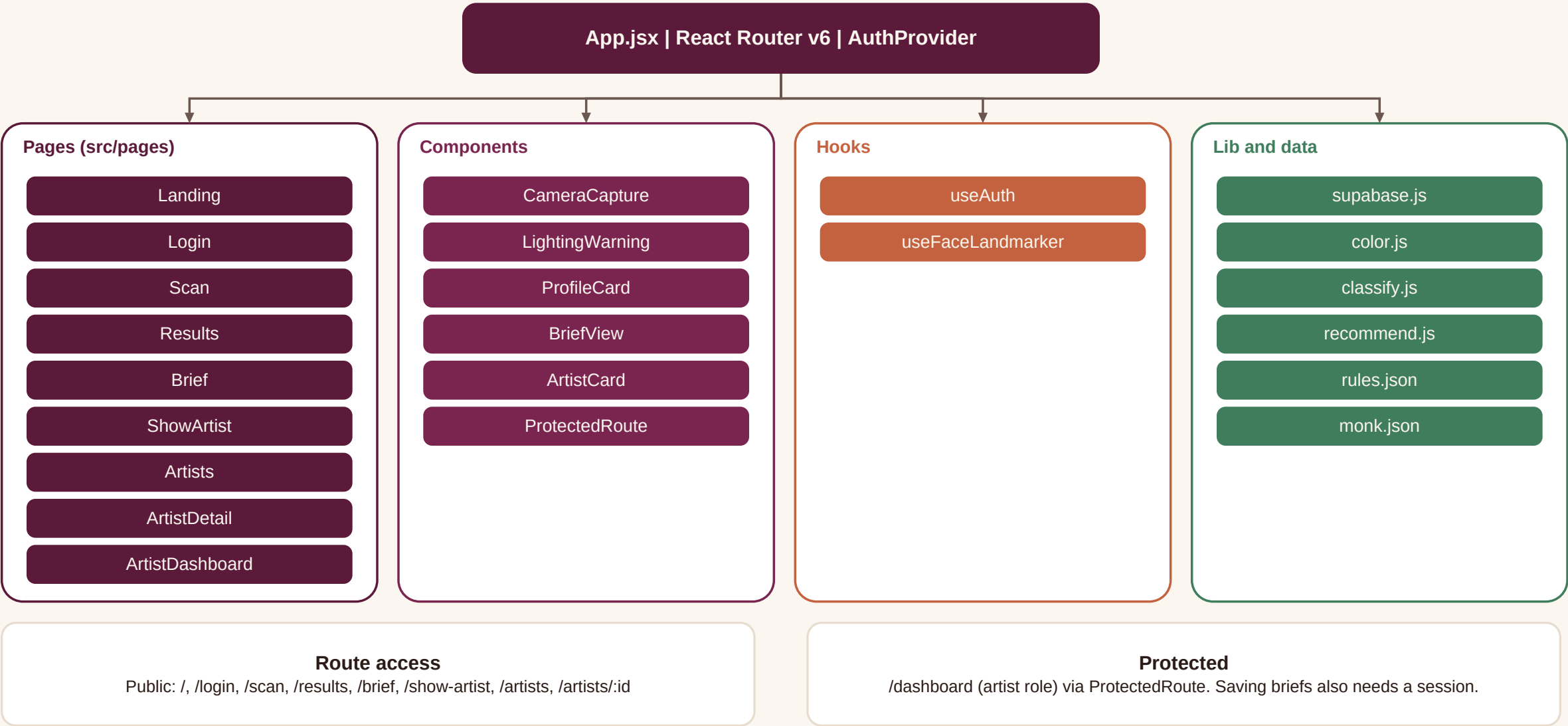
axe DevTools

Figma (free)

Free-tier limits change over time. Confirm current Supabase and Vercel limits before launch.

Frontend Architecture

Vite + React SPA. State via hooks and Context. Vision model lazy-loaded on the Scan route only



Computer Vision Pipeline

Six stages, all client-side. The captured frame never leaves memory

1 Capture

Front camera frame drawn to a canvas. Upload fallback if permission is denied.

```
getUserMedia({video:{facingMode:'user'}})
```

2 Detect

One face, still-image mode. Returns 478 normalized landmarks.

```
FaceLandmarker.detect(canvas)
```

3 Sample

Three 12x12 px patches: left cheek, right cheek, forehead. Eyes, lips, brows and hair avoided.

```
landmarks 50, 280, 151
```

6 Classify

Nearest Monk swatch in LAB gives depth (plus runner-up). Hue angle gives undertone.

```
classifyDepth(), classifyUndertone()
```

5 Convert

sRGB to linear to XYZ (D65) to CIELAB. Patch L* values kept for the lighting check.

```
rgbToLab([r,g,b])
```

4 Robust average

Per-channel median in each patch rejects highlights and blemishes. Mean of the three patches.

```
median(R), median(G), median(B)
```

Landmark indices to be verified with a debug overlay

Color Science

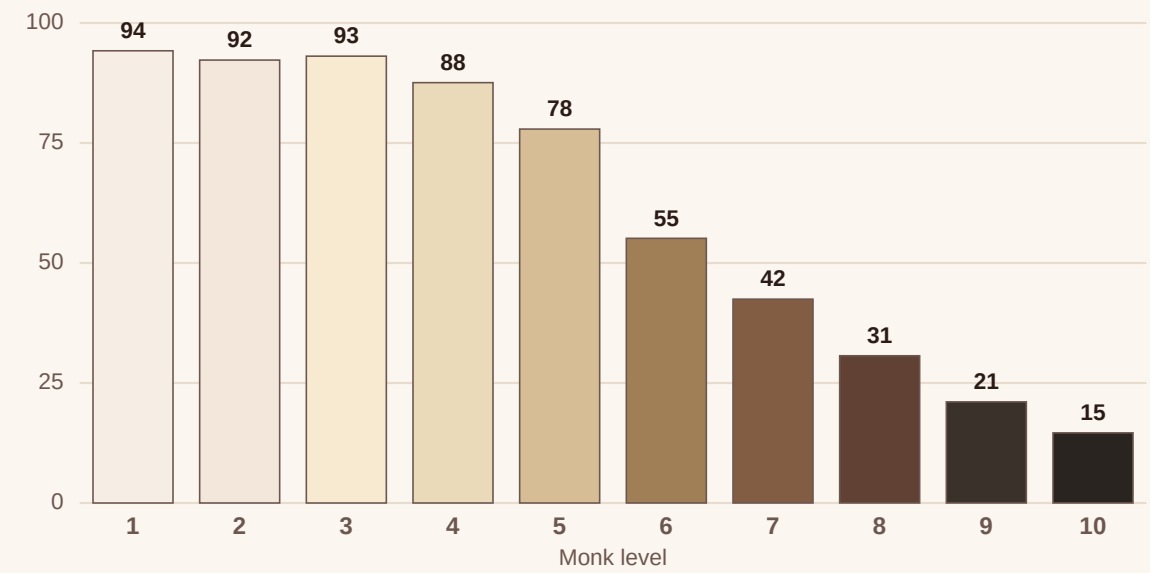
CIELAB separates lightness from color, which makes depth and undertone measurable



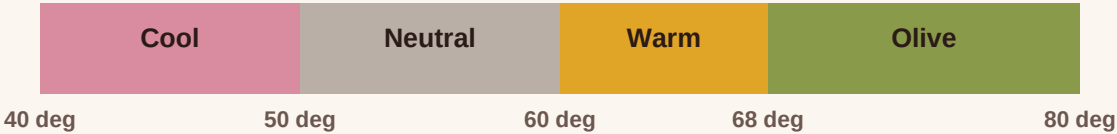
Formulas used

lin(c) = c/12.92 if c <= 0.04045
= ((c+0.055)/1.055)^2.4 otherwise
ITA = atan((L* - 50) / b*) * 180/pi
hue = atan2(b*, a*) * 180/pi

Measured L* of the 10 Monk reference swatches



Undertone by hue angle (starting thresholds)



Olive is confirmed with a low a* value (yellow-green lean). If the three patches disagree, confidence is lowered and the user can adjust manually.

Classification logic

// depth: nearest Monk swatch in LAB
level = argmin_s sqrt((L-Ls)^2 + (a-as)^2 + (b-bs)^2)
range = [level, second_nearest]

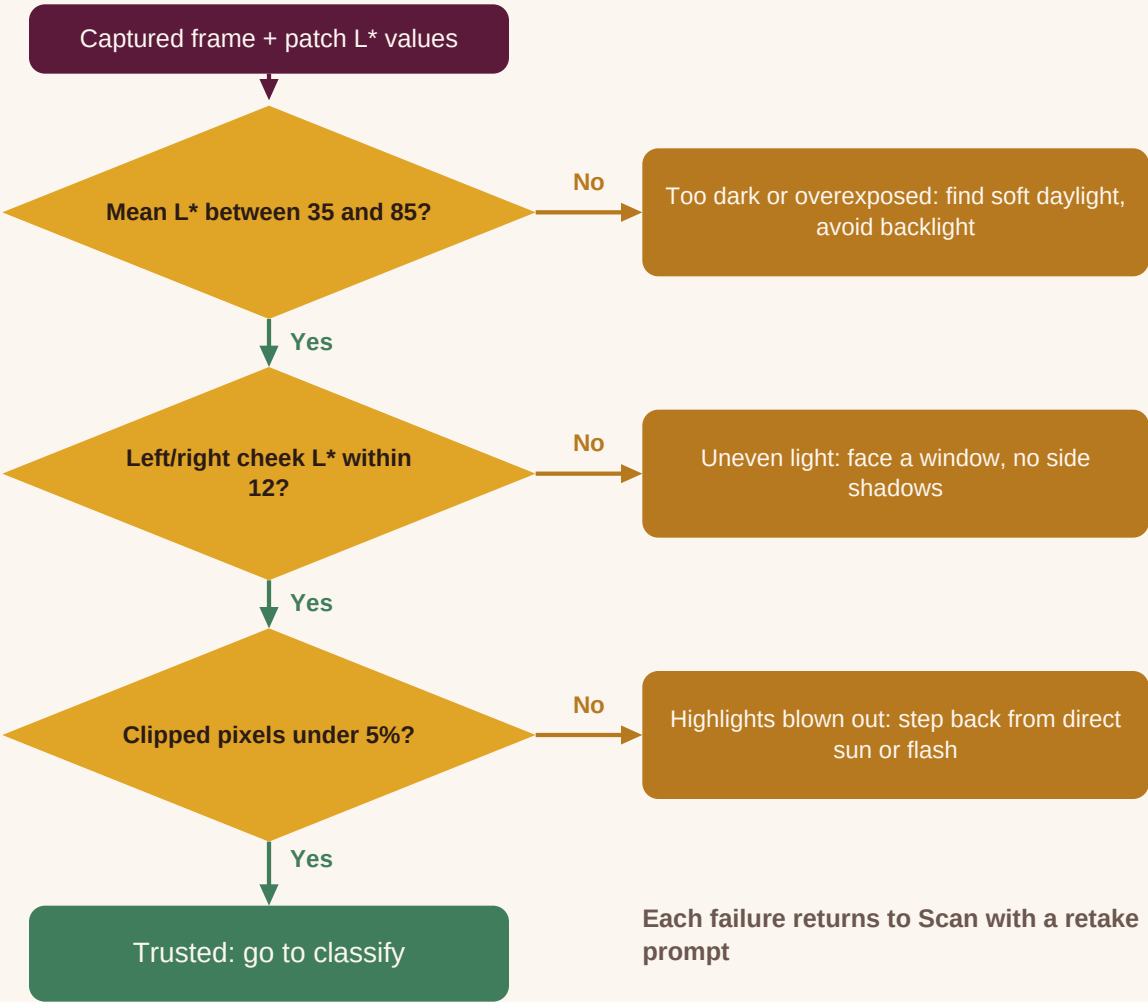
// rule key
group = fair | light-medium | tan | deep
key = group + '-' + undertone // e.g. deep-warm

Calibration note

Thresholds are a starting point. Calibrate on 100+ consented samples labelled by professional makeup artists, then tune per depth group. Phone beauty filters and auto white balance can shift color, so results are shown as a range.

Lighting Quality Gate

Bad light gives bad color. Three checks decide whether a scan is trusted



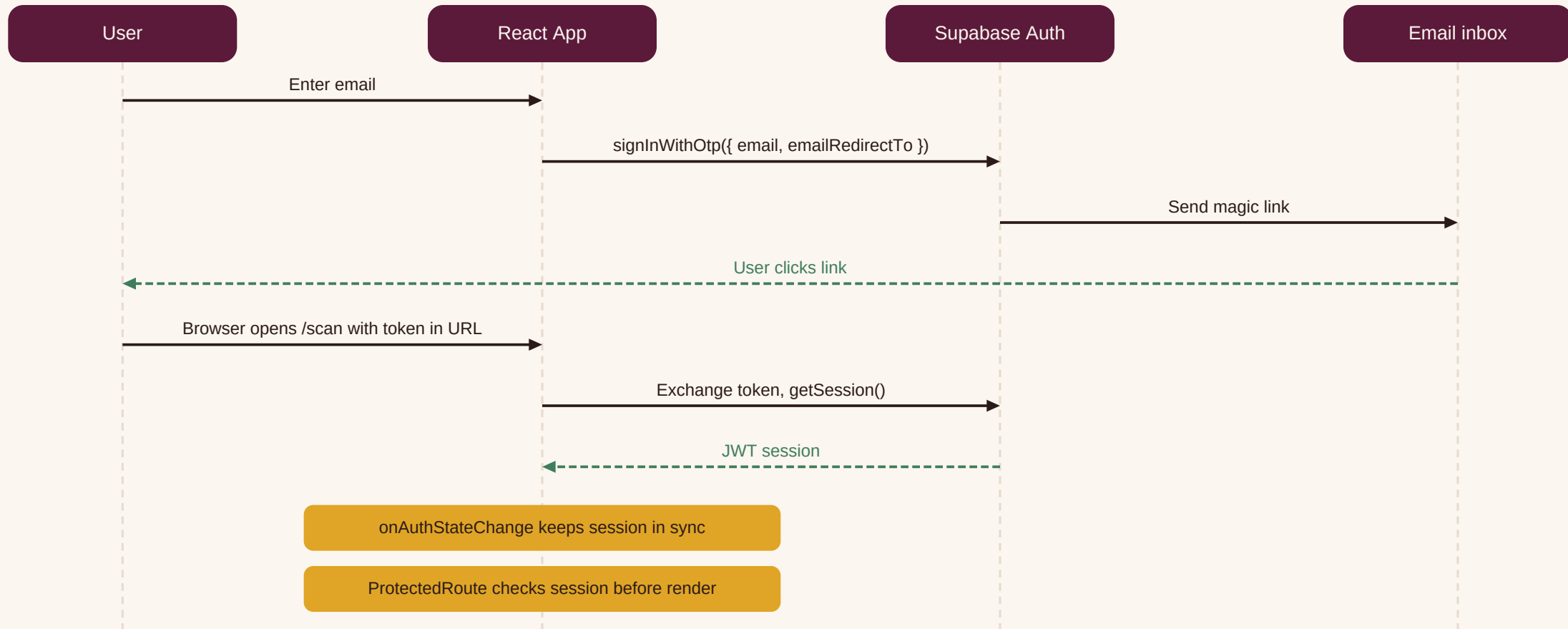
Check	Rule (tunable)	Purpose
Exposure	Mean L* 35 to 85	Reject dark or washed-out frames
Evenness	$\text{abs}(L^* \text{ left} - L^* \text{ right}) \leq 12$	Detect side lighting
Clipping	< 5% clipped pixels	Avoid blown highlights
Color cast	White-card reference (optional): normalize by card RGB	Correct warm indoor bulbs

Guidance shown on the Scan screen

Ask for natural window light, face toward the light, no filters, no heavy makeup. The result screen always shows a depth range, and the bride can nudge depth and undertone manually.

Authentication Flow (Supabase Magic Link)

Passwordless email login; scanning works without an account



Setup

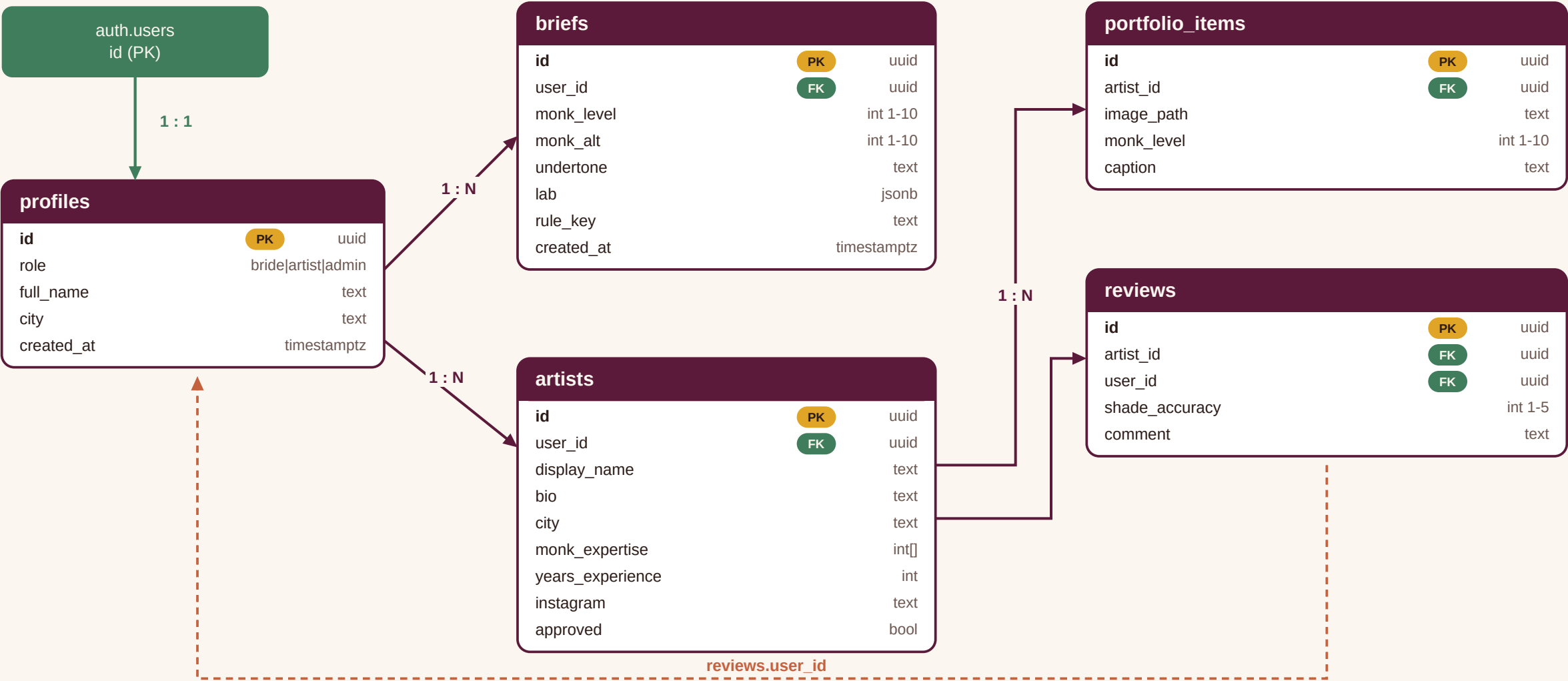
Whitelist `http://localhost:5173` and the Vercel URL under Authentication > URL Configuration.

Guest vs signed in

Guests can scan and view the brief. A session is required to save briefs, share, or review.

Database Schema (Postgres on Supabase)

5 tables in public, keyed to Supabase Auth users; row-level security on every table



Indexes: GIN on artists.monk_expertise, btree on artists.city, portfolio_items(artist_id, monk_level). Selfie images are never stored.

Privacy and Security

The photo stays on the phone. Only derived numbers and account data reach the cloud

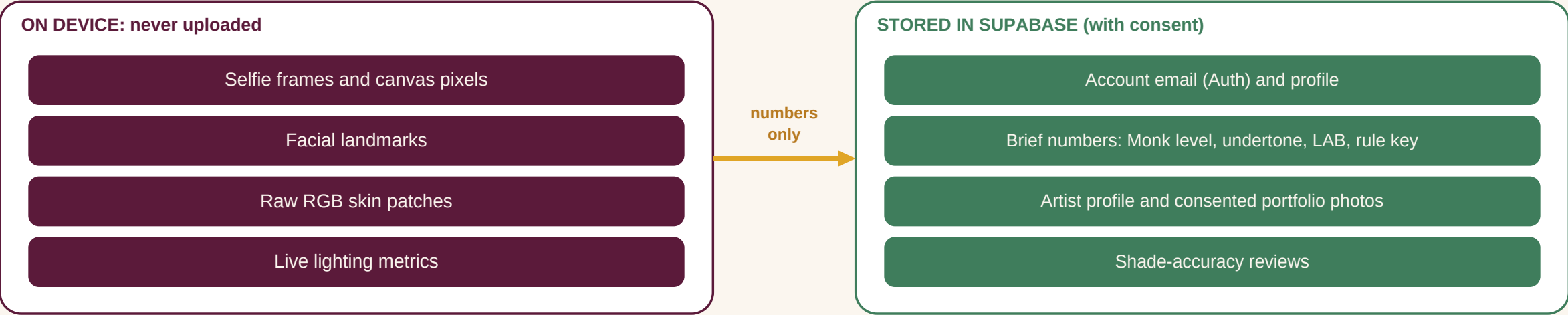
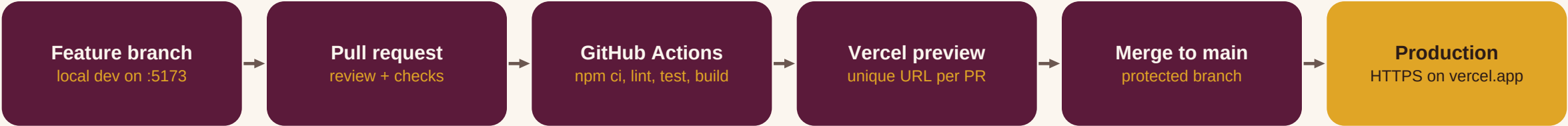


Table / bucket	Read	Write
profiles	Own row	Own row
briefs	Own rows only	Own rows only (user_id = auth.uid())
artists	Public if approved; owner sees own	Owner insert and update
portfolio_items	Public for approved artists	Owner of the artist record
reviews	Public	Author only; one review per artist per user
storage: portfolios	Public read	Authenticated, only in own-uid folder; JPEG/PNG/WebP, max 5 MB

Anon key is safe in the client only because RLS is enabled. The service_role key is never shipped to the browser.

Build, Deploy and Performance

Push to GitHub, verify, preview, promote



Environment variables

```
VITE_SUPABASE_URL=https://<project>.supabase.co
VITE_SUPABASE_ANON_KEY=<anon key>

# set in Vercel > Settings > Environment Variables
# never commit .env.local
```

Bootstrap

```
npm create vite@latest skinsync -- --template react
npm i @supabase/supabase-js react-router-dom
npm i @mediapipe/tasks-vision lucide-react jspdf
npm i -D tailwindcss @tailwindcss/vite vitest
```

Performance budget

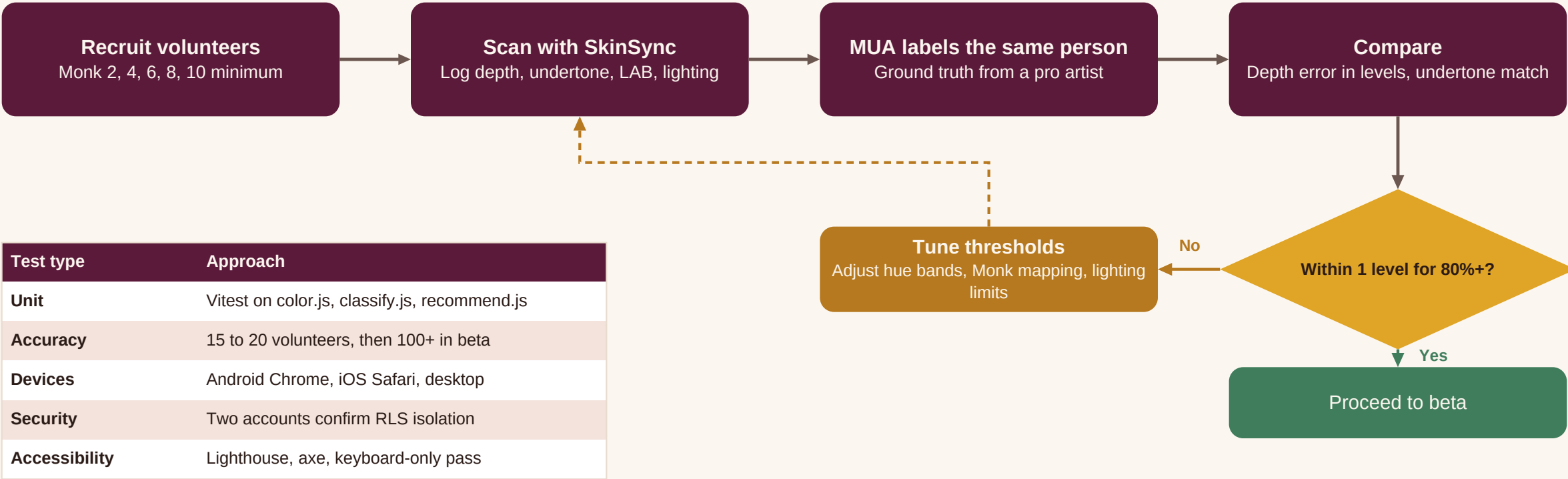
Metric	Target
First-load JS (excluding model)	< 250 KB gzip
MediaPipe model	Lazy-load on Scan page only
Time to interactive (4G)	< 4 s
Scan compute after capture	< 1 s
Uploaded images	Resize to max 1200 px; lazy-load lists
Lighthouse mobile performance	> 80

Demo-day tip

Self-host the MediaPipe wasm and .task model in /public if the CDN is unreliable on the demo network.

Testing and Calibration Loop

Accuracy across the whole skin range is the product, so it is tested first



Pitch tip

Show measured accuracy across the Monk range in the demo. If a tone group performs worse, say so and show the fix. Honesty about limits builds more trust than a perfect-looking number.

Technical Risks and What Comes Next

Known limits, planned mitigations and the technical roadmap

Risk	Mitigation
Beauty filters, auto white balance alter color	Advise no filters; white-card calibration; show ranges
Undertone is subjective	Show confidence; user can confirm or adjust
MediaPipe CDN outage	Self-host wasm and model in /public
Supabase free project pauses when idle	Keep active before demo; keep SQL exportable
iOS Safari camera quirks	playsinline, early device tests, upload fallback
Classifier bias toward lighter skin	Test across Monk 1-10; publish accuracy; tune per group

Technical roadmap

Edge Function
Secure share links for briefs

Rate limiting
Abuse controls on writes

Calibration set
Per-device color profiles

Lightweight ML
Classifier trained on labelled samples

i18n
react-i18next: Hindi first

AR try-on
Face-mesh shade overlays

SkinSync

Your skin. Your wedding. Enhanced, not erased.



Thank you | Questions?

On-device vision. Honest accuracy. Free to build, ready to grow.